JeongBeom Baek (백정범)

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Education

Sungkyul University (성결대학교), Computer Engineering — Anyang, South Korea, 2026 – present

- 1st Semester GPA: 3.97 / 4.5 (16 credits)
- C Programming: A+ (98) | AI Understanding & Application: A+ (98)
- Basic Writing: A+ (95) | College Mathematics: B+ (87)
- Research & Teaching Assistant under Prof. Hyeji Yang
- Nominated for university curriculum development initiative

Incheon Youngsun High School (인천영선고등학교), General Studies (Science Track) — Incheon, South Korea, 2023 – 2025

- Founded CodingLab (코딩랩) — designed and taught a deep learning theory curriculum
- Information/CS: Rank 2 (consistently top performer across 3 years)
- Programming (elective): A grade — implemented NLP tokenization algorithms and Llama 3.2 Q&A pipeline
- AI & Physical Computing: Researched on-device AI chips, sLLM, and knowledge distillation to address LLM power consumption
- Mathematics & Statistics: Derived backpropagation equations via chain rule/partial derivatives and analyzed AI output probability distributions
- Social Issues: Authored a report on AI hallucinations, dependence risks, and the necessity of AI literacy
- Data Analysis: Utilized Orange3 for data mining and GeoGebra for complex mathematical visualizations
- KSEF 2024: Published research on improving LLM reasoning via critical thinking frameworks

Experience

Sungkyul University, Undergraduate Student & Independent Researcher — Anyang, South Korea, Mar 2026 – present, 5 months

Concurrently pursuing studies and participating in departmental initiatives. Independently archived and analyzed over 40 AI agent papers for an academic course assignment.

- Exceeded academic assignment requirements by analyzing 40+ international papers on AI agents and RL
- Nominated to participate in a university curriculum development project
- Discussed Nunchi (눈치) humor-weight algorithm concept with Prof. Seong-jun Park (Dept. Head), resulting in an encouraging exploration of human-robot interaction

CodingLab (코딩랩), Lead Instructor & Re-founder — Incheon Youngsun High School, Mar 2025 – Dec 2025, 10 months

When the previous programming club dissolved, I stepped up to recreate the club as CodingLab. Designed a deep learning theory curriculum from MIT 6.S191 and Hands-On ML materials. Delivered ~1 semester of lectures to students with no prior ML background.

- Designed 7-week DL curriculum: perceptrons, backpropagation, CNNs, RNNs, Transformers, attention mechanisms
- Created original lecture materials (80+ slides) with mathematical rigor — equations, diagrams, code examples
- Mentored students from zero coding knowledge to understanding neural network architectures

Incheon Youngsun High School, Coding Club Member & TA (코딩파티) — Incheon, South Korea
Mar 2023 – Dec 2024
1 year 10 months

Active member of the school coding club. Helped peers with Python programming, developed apps using App Inventor, and served as a technical resource for classmates.

- Built custom shoe and eco-friendly fashion coordination apps using App Inventor, incorporating recommendation algorithms
- Coded a classroom seat arrangement program used by the entire class to solve real-world scheduling issues
- Tutored peers in Python (loops, functions, game programs) and served as a technical resource for classmates

Publications

Improving LLM Language Understanding through the Application of Critical Thinking — 2024

Proposed applying human critical thinking patterns to LLM inference — generating and verifying logical hypotheses before producing responses, rather than direct token prediction. Fine-tuned Llama 3.2 3B with LoRA on a custom Korean dataset (merged-v2). The critical thinking model scored 0.566 vs baseline 0.466 on language understanding, reasoning, and requirement fulfillment metrics.
JeongBeom Baek
ericbaek0828-netizen.github.io/assets/pdf/KSEF_2024_Paper.pdf

Awards

2026 Department Game Jam — High Commendation — July 2026

Built 'Escape from Uniformity' — a 2D side-scrolling adventure with suspicion gauge, stat-based NPC persuasion, and multi-ending structure. Used Harness Engineering (AI-assisted no-code workflow) as a freshman with a 3-person team over 4 days. Received highly positive verbal feedback for development execution.
Sungkyul University, Computer Engineering Dept.

Skills

Programming & Engineering

AI & Machine Learning

Research & Teaching

Tools & Platforms

Languages

Korean
Native speaker

English
Professional communication

Interests

AI & Systems

Knowledge Management

Cognitive Science

Projects

J.A.R.V.I.S. 2026 – present

Conceptual framework and architecture in active planning for a multi-agent orchestration system, focusing on layered agent routing, state management, and the Nunchi social context engine.

- Designing layered agent routing and automated quality gates for multi-agent workflows
- Nunchi algorithm: humor/sarcasm weight filtering based on unconscious physical signals
- Currently focusing on robust architectural planning and component design

Escape from Uniformity (획일화 속 탈출) June 2026 – July 2026

2026 Department Game Jam entry. A 2D side-scrolling adventure satirizing conformist society, with suspicion gauge, stat-based NPC persuasion, and multi-ending narrative structure.

- Harness Engineering workflow — AI-assisted no-code development as a freshman
- Perfect score in development category; judges called me 'a developer to the bone'

Critical Thinking LLM (KSEF 2024) June 2024 – Nov 2024

Research project applying human critical thinking patterns to LLM inference. Fine-tuned Llama 3.2 3B to generate and verify hypotheses before responding.

- Custom Korean dataset (merged-v2) published on Hugging Face
- 21% improvement in language understanding score over baseline Llama 3.2